Συστήματα Διαχείρισης και Ανάλυσης Δεδομένων
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Δεύτερη Εργασία

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ΖΗΤΗΜΑ 1

```
-- Load data from text file

CREATE TABLE CardsTransactions
(pid int,
 pname varchar(50),
 age int,
 gender char (1),
 cardno char(16),
 card_brand varchar(30),
 card_type varchar(20),
 tdate DATETIME,
 amount DECIMAL(6,2),
 ttc int,
 trans_type varchar(20),
 mcc int,
 merchant_city varchar(50)
 );

BULK INSERT CardsTransactions
FROM 'C:\MY_DATA\cardsTransactions.txt'
WITH (FIRSTROW =2, FIELDTERMINATOR='|', ROWTERMINATOR = '\n');

-- Create star schema

CREATE TABLE persons
(pid int primary key,
 pname varchar(50),
 age int,
 gender char(1)
 );

 insert into persons
   select distinct pid,pname,age,gender from CardsTransactions


 CREATE TABLE cards
 (cardno CHAR(16) primary key,
  card_brand varchar(30),
  card_type varchar(20)
  );

 insert into cards
    select distinct cardno,card_brand, card_type from CardsTransactions

 CREATE TABLE transaction_types
 (ttc int primary key,
  trans_type varchar(20)
  );
```

```
 insert into transaction_types
    select distinct ttc,trans_type from CardsTransactions

 CREATE TABLE cities
 (mcc int primary key,
  merchant_city varchar(50),
  );

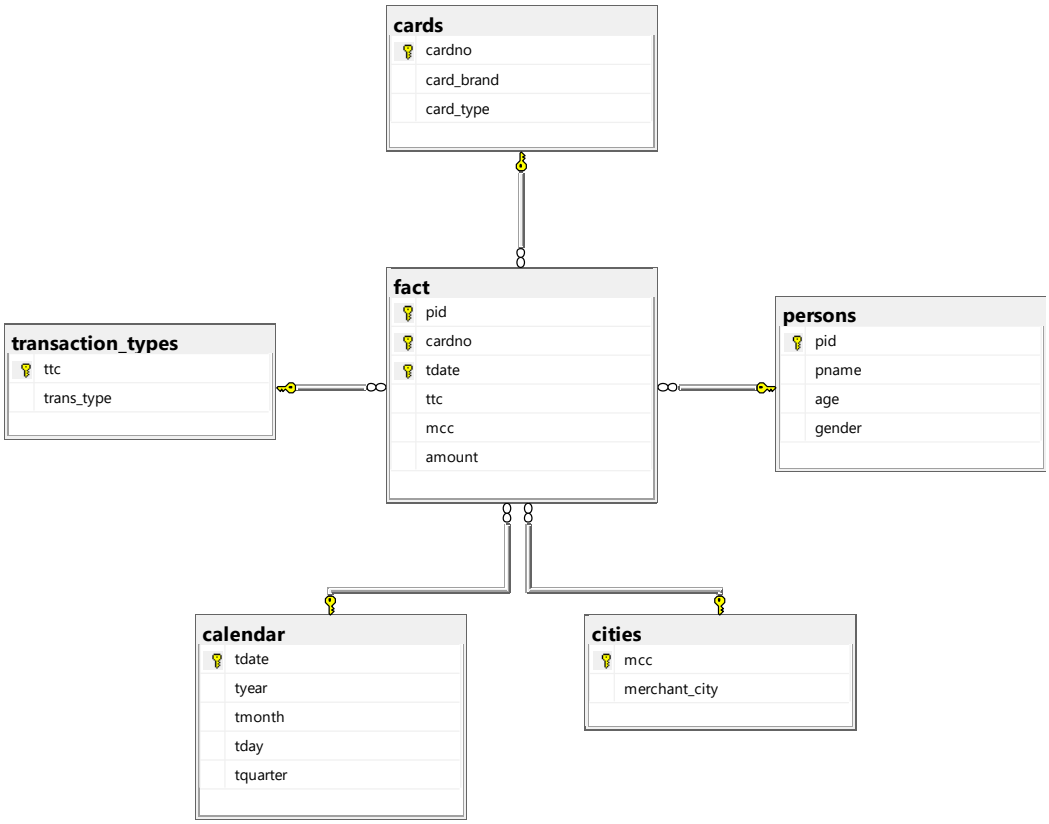
 insert into cities
   select distinct mcc,merchant_city from CardsTransactions

drop table calendar
 CREATE TABLE calendar
 (tdate DATETIME primary key,
  tyear int,
  tmonth int,
  tday int,
  tquarter int
  );

  set datefirst 1;
  insert into calendar
  select distinct tdate, datepart(year,tdate), datepart(month,tdate),
datepart(day,tdate), datepart(quarter, tdate)
    from CardsTransactions

CREATE TABLE fact
 (pid int foreign key references persons(pid),
  cardno char(16) foreign key references cards(cardno),
  tdate DATETIME foreign key references calendar(tdate),
  ttc int foreign key references transaction_types(ttc),
  mcc int foreign key references cities(mcc),
  amount DECIMAL(6,2),
 PRIMARY KEY (pid, cardno, tdate)
 );

 insert into fact
 select distinct pid,cardno,tdate,ttc,mcc,sum(amount)
  from cardsTransactions
  group by pid,cardno,tdate,ttc,mcc
```


cards
cardno
card_brand
card_type
fact
pid
cardno
tdate
ttc
mcc
amount
transaction_types
ttc
trans_type
persons
pid
pname
age
gender
calendar
tdate
tyear
tmonth
tday
tquarter
cities
mcc
merchant_city

ΖΗΤΗΜΑ 2

```
--1. Αξία συναλλαγών ανα πόλη
select merchant_city, sum(amount)
 from cities, fact
where cities.mcc = fact.mcc
group by merchant_city
order by merchant_city

--2. Αξία συναλλαγών ανα έτος και φύλο

select tyear, gender, SUM(amount)
 from calendar,persons, fact
 where calendar.tdate=fact.tdate and
       persons.pid=fact.pid
          group by tyear, gender
order by tyear desc

--3. Αριθμός και Αξία συναλαλγών ανα επωνυμία και είδος κάρτας

select card_brand, card_type, COUNT(*), SUM(amount)
  from cards, fact
where cards.cardno=fact.cardno
group by card_brand, card_type

--4 Αξία συναλλαγών ανα τύπο συναλλαγής για κάθε τρίμηνο του έτους 2019
select trans_type, tquarter, sum(amount)
  from transaction_types, fact, calendar
where transaction_types.ttc=fact.ttc and
      fact.tdate=calendar.tdate and tyear=2019
         group by trans_type, tquarter

--5 Αριθμός και αξία ONLINE συναλαλγών ανα Ετος, φύλο και Ηλικία
    select tyear, gender, age, SUM(amount)
        from  persons, calendar, transaction_types, fact
        where persons.pid=fact.pid and
              calendar.tdate=fact.tdate and
                 transaction_types.ttc=fact.ttc and trans_type='Online
Transaction'
                 group by ROLLUP(tyear,gender,age)
                 order by tyear, gender, age
```

ZHTHMA 3

```
--6 CUBE

select tyear, card_brand, gender, count(*)
   from calendar,cards, persons, fact
where calendar.tdate=fact.tdate and
      cards.cardno=fact.cardno and
      persons.pid=fact.pid
         group by CUBE (tyear, card_brand, gender)


create view v1
with schemabinding as
select C.tyear, CA.card_brand,  P.gender, count_big(*)  as Num_of_trans
   from dbo.calendar as C,dbo.cards as CA, dbo.persons as P, dbo.fact as F
where C.tdate=F.tdate and
      CA.cardno=F.cardno and
      P.pid=F.pid
         group by C.tyear, CA.card_brand, P.gender

create unique clustered index idx1 on V1 (tyear,card_brand,gender)



select tyear, card_brand, sum (Num_of_trans)
 from v1
group by tyear, card_brand

select tyear, gender, sum (Num_of_trans)
 from v1
group by tyear, gender

select card_brand, gender, sum (Num_of_trans)
 from v1
group by card_brand, gender

select tyear, sum (Num_of_trans)
 from v1
group by tyear

select card_brand, sum (Num_of_trans)
 from v1
group by card_brand

select gender , sum (Num_of_trans)
 from v1
group by gender

select  sum (Num_of_trans)
 from v1
```